

Newtonian Mechanics

Analytical Mechanics

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Definition

*The **zeroth law of dynamics** states that*

- a) Our space is three-dimensional and Euclidean, and time is one-dimensional.*
- b) **(Galileo's principle of relativity)** There exist reference systems, called **inertial**, possessing the following two properties:*
 - b'1) all the laws of nature at all moments of time are the same in all inertial reference frames.*
 - b'2) all the reference systems in uniform rectilinear motion with respect to an inertial one are themselves inertial.*
- c) **(Newton's principle of determinacy)** The initial state of a mechanical system (the totality of positions and momentums at some moment in time) uniquely determines all of its motion.*



Isaac Newton
1642-1727

Historical Notes

Isaac Newton (1642-1727) was born in Woolsthorpe, in Lincolnshire, England, on December 25th; his father, a farmer, had died three months earlier. His mother remarried when he was three years old and he was sent to live with his grandmother. He returned to his mother at age eleven, only to be sent to boarding school in Grantham the next year. Fortunately, a perceptive teacher noticed his mathematical talent and, in 1661, Newton entered Trinity College at Cambridge University, where he studied with Isaac Barrow. When the bubonic plague struck in 1665-1666, leaving dead nearly 70,000 persons in London, the university closed and Newton spent two years back in Woolsthorpe. It was during this period that he formulated his basic ideas concerning optics, gravitation, and his method of “fluxions”, later called “calculus”. He returned to Cambridge in 1667 and was appointed Lucasian professor in 1669. His theories of universal gravitation and planetary motion were published to world acclaim in 1687 under the title *Philosophiæ Naturalis Principia Mathematica*. However, he neglected to publish his method of inverse tangents for finding areas and other work in calculus, and this led to a controversy over priority with Leibniz. Following an illness, he retired from Cambridge University and in 1696 was appointed Warden of the British mint. However, he maintained contact with advances in science and mathematics and served as President of the Royal Society from 1703 until his death in 1727. At his funeral, Newton eulogised as “the greatest genius that ever existed”. His place of burial in Westminster Abbey is a popular tourist site.

The following definitions and laws are directly extracted from Newton's book *Philosophiæ Naturalis Principia Mathematica* (1687) which was translated in 1729 by Andrew Motte.

Definition I

The quantity of matter is the measure of the same, arising from its density and bulk conjunctly.

Definition II

The quantity of motion is the measure of the same, arising from the velocity and quantity of matter conjunctly.

Definition III

The vis insita, or innate force of matter, is a power of resisting, by which every body, as much as in it lies, endeavours to preserve in its present state, whether it be of rest, or moving uniformly forward in a right line.

Definition IV

An impressed force is an action exerted upon a body, in order to change its state, either of rest, or of moving uniformly forward in a right line.

Definition V

A centripetal force is that by which bodies are drawn or impelled, or any way tend, towards a point as to a centre.

Definition VI

The absolute quantity of a centripetal force is the measure of the same proportional to the efficacy of the cause that propagates it from the centre, through the spaces round about.

Definition VII

The accelerative quantity of a centripetal force is the measure of the same, proportional to the velocity which it generates in a given time.

Definition VIII

The motive quantity of a centripetal force, is the measure of the same, proportional to the motion which it generates in a given time.

AXIOMS, OR LAWS OF MOTION

Law I (Newton's First Law)

Every body perseveres in its state of rest, or of uniform in a right line, unless it is compelled to change that state by forces impressed thereon.

Law II (Newton's Second Law)

The alteration of motion is ever proportional to the motive force impressed; and is made in the direction of the right line in which that force is impressed.

Law III (Newton's Third Law)

To every action there is always opposed an equal reaction; or the mutual actions of two bodies upon each other are always equal, and directed to contrary parts.

Definition

A **law** is a descriptive generalisation about how some aspect of the natural world behaves under stated circumstances.

In science, a **theory** is a well-substantiated explanation of some aspect of the natural world that can incorporate facts, laws, inferences, and tested hypotheses.

Definition

A **reference frame** is an ordered quadruple $\mathcal{R} = (O, \mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3)$ where $\{\mathbf{n}_i\}_{i=1}^3$ is a linearly independent set of vectors and O is a point, called the **origin**. A reference frame $\mathcal{R} = (O, \mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3)$ is said to be **orthonormal** if

$$\mathbf{n}_i \cdot \mathbf{n}_j \equiv \langle \mathbf{n}_i, \mathbf{n}_j \rangle = \delta_{ij}$$

where δ_{ij} denotes Kronecker's delta and $\langle \cdot, \cdot \rangle$ the Euclidean inner product. In this case we usually write $\mathcal{R} = (O, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ to denote the fact that $\mathcal{R}$ is orthonormal.

Definition

Moreover, the reference frame is said to be **right-handed** if

$$\mathbf{e}_i \times \mathbf{e}_j = \varepsilon_{ijk} \mathbf{e}_k,$$

with $\times$ a bilinear product defined by the previous equation which turns $\mathbb{R}^3$ into a nonassociative, noncommutative algebra.

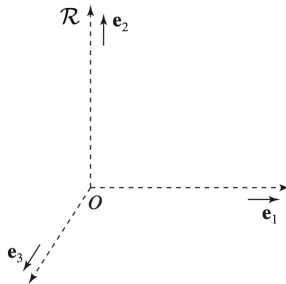


Figure 1: An orthonormal and right-handed reference frame $\mathcal{R}$

Definition

Given any reference frame $\mathcal{R}$, the elements $B = \{\mathbf{e}_i\}_{i=1}^3$ are known to form a basis for $\mathbb{R}^3$, so that given any vector $\mathbf{r}$ we can express it as $\mathbf{r} = r_i \mathbf{e}_i$. We write

$$\mathbf{r} = \begin{pmatrix} r_1 \\ r_2 \\ r_3 \end{pmatrix}_{\mathcal{R}}$$

and say that the column vector $\mathbf{r}$ represents $\mathbf{r}$ in $\mathcal{R}$.

Definition

The angle θ between two nonzero vectors $\mathbf{u}, \mathbf{v}$ is defined via

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{||\mathbf{u}|| ||\mathbf{v}||}.$$

Proposition 1.1

Let θ be the angle between two nonzero vectors $\mathbf{u}, \mathbf{v}$. Then

$$\mathbf{u} \times \mathbf{v} = ||\mathbf{u}|| ||\mathbf{v}|| \sin \theta \mathbf{n}$$

where $\mathbf{n}$ is the unit vector perpendicular (in the right hand sense) to the plane containing $\mathbf{u}$ and $\mathbf{v}$.

Definition

Let $\mathbf{r} : I \subseteq \mathbb{R}^+ \cup \{0\} \rightarrow \mathbb{R}^3$ and let $\mathcal{R}$ be a reference frame. If the scalar functions r_i , defined by $\mathbf{r}(t) = r_i(t)\mathbf{e}_i$, are all differentiable at $t_0 \in I$, then $\mathbf{r}$ is said to be **differentiable in the reference frame $\mathcal{R}$** at t_0 . If $\mathbf{r}$ is differentiable at each point of its domain in a given reference frame, we will simply say that $\mathbf{r}$ is differentiable. Furthermore, the **derivative operator in $\mathcal{R}$** is defined as a linear mapping ${}^{\mathcal{R}}\mathrm{d}/\mathrm{d}t$ acting on differentiable vector functions, satisfying the Leibniz rule with respect to the multiplication by a scalar, such that

$$\left. {}^{\mathcal{R}}\frac{\mathrm{d}\mathbf{e}_1}{\mathrm{d}t} \right|_{t_0} = \left. {}^{\mathcal{R}}\frac{\mathrm{d}\mathbf{e}_2}{\mathrm{d}t} \right|_{t_0} = \left. {}^{\mathcal{R}}\frac{\mathrm{d}\mathbf{e}_3}{\mathrm{d}t} \right|_{t_0} = 0 \quad \text{for all } t_0 \in I,$$

and such that

$$\left. {}^{\mathcal{R}}\frac{\mathrm{d}f(t)}{\mathrm{d}t} \right|_{t_0} = f'(t_0) \equiv \left. \frac{\mathrm{d}f}{\mathrm{d}t} \right|_{t_0}$$

with $f'(t_0)$ the usual scalar derivative.

Theorem 1.2 (Differentiation Rules)

Let $\mathbf{r}, \mathbf{s} : I \rightarrow \mathbb{R}^3$ be differentiable vector functions and let $f : \mathbb{R} \rightarrow I$, $g : \mathbf{r}(I) \rightarrow \mathbb{R}$ be a differentiable scalar functions. Then for any orthonormal and right-handed reference frame $\mathcal{R}$ we have

i) (Leibniz rule with respect to the Euclidean inner product)

$$\frac{d}{dt} (\mathbf{r} \cdot \mathbf{s}) = \frac{d\mathbf{r}}{dt} \cdot \mathbf{s} + \mathbf{r} \cdot \frac{d\mathbf{s}}{dt}$$

ii) (Leibniz rule with respect to the cross product)

$$\frac{d}{dt} (\mathbf{r} \times \mathbf{s}) = \frac{d\mathbf{r}}{dt} \times \mathbf{s} + \mathbf{r} \times \frac{d\mathbf{s}}{dt}$$

iii) (Chain rule for vector differentiation)

$$\frac{d}{d\tau} (\mathbf{r} \circ f) = \left(\frac{d\mathbf{r}}{dt} \circ f \right) \frac{df}{d\tau} \quad \text{and} \quad \frac{d}{dt} (g \circ \mathbf{r}) = \nabla_{\mathcal{R}} g \cdot \frac{d\mathbf{r}}{dt} + \frac{\partial g}{\partial t}.$$

Definition

Given any differentiable mapping $\mathbf{r} : I \rightarrow \mathbb{R}^3$ with I a real interval. If $\mathcal{R}$ is a reference frame, then for any $t \in I$ let

$$P = \begin{pmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{pmatrix}_{\mathcal{R}}$$

represent the vector $\mathbf{r}(t)$ in $\mathcal{R}$. Then $\mathbf{r}(t)$, now written as $\mathbf{r}_{P/O}$ is called the **position vector** of the point P relative to the origin O of $\mathcal{R}$.

Definition

The derivative

$$\frac{{}^{\mathcal{R}}d\mathbf{r}_{P/O}(t_0)}{dt} \equiv {}^{\mathcal{R}}\mathbf{v}_{P/O}(t_0)$$

*is called the **velocity vector** of point P at instant t_0 relative to frame $\mathcal{R}$.
Moreover, the second derivative*

$$\frac{{}^{\mathcal{R}}d^2\mathbf{r}_{P/O}(t_0)}{dt^2} \equiv {}^{\mathcal{R}}\mathbf{a}_{P/O}(t_0)$$

*is called the **acceleration vector** of point P at instant t_0 relative to frame $\mathcal{R}$.*

Definition

The **linear momentum** $\mathbf{p}$ of a particle P is defined as the product of the particle inertial mass and its velocity:

$${}^{\mathcal{R}}\mathbf{p}_{P/O} = m_P {}^{\mathcal{R}}\mathbf{v}_{P/O}.$$

Definition

*A reference frame is called an **inertial frame** if Newton's laws are indeed valid in that frame. We often use the symbol $\mathcal{I}$ to label an inertial reference frame.*

Proposition 1.3

If Newton's laws are valid in one reference frame, then they are also valid in any reference frame in uniform motion with respect to the first system.

Definition

*The previous result is called **Galilean invariance** or the **principle of Galilean relativity**.*

Definition

The set $\text{Gal}(3)$ of **Galilean transformations** is given by:

- **(Spatial translations)** $\mathbf{r}'(t) = \mathbf{r}(t) + \mathbf{r}_0$, $\mathbf{r}_0 \in \mathbb{R}^3$.
- **(Time translations)** $\mathbf{r}'(t) = \mathbf{r}(t + t_0)$, $t_0 \in \mathbb{R}^+$.
- **(Galilean boosts)** $\mathbf{r}'(t) = \mathbf{r}(t) - \mathbf{v}t$, $\mathbf{v} \in \mathbb{R}^3$.
- **(Rotations)** $\mathbf{r}'(t) = \mathbf{R}\mathbf{r}(t)$, $\mathbf{R} \in SO(3)$, $\mathbf{R} \neq \mathbf{R}(t)$.

A general element of $\text{Gal}(3)$ is written as $g = (\mathbf{R}, \mathbf{v}, \mathbf{x}_0, t_0)$ which acts on $(\mathbf{x}, t)$ as $g(\mathbf{x}, t) = (\mathbf{R}\mathbf{x} + \mathbf{v}t + \mathbf{x}_0, t + t_0)$.

Proposition 1.4

*Let $\circ$ be the composition of transformations in $\text{Gal}(3)$. Then $(\text{Gal}(3), \circ)$ forms a group, called the **Galilean group**.*

Remark

Newton's second law states that time derivative with respect to the inertial frame of the linear momentum of a point mass P is equal to the total force $\mathbf{F}_P$ acting on P , in symbols,

$$\mathbf{F}_P = \frac{{}^I d}{{}^I dt} ({}^I \mathbf{p}_{P/O}) = m_P {}^I \mathbf{a}_{P/O}.$$

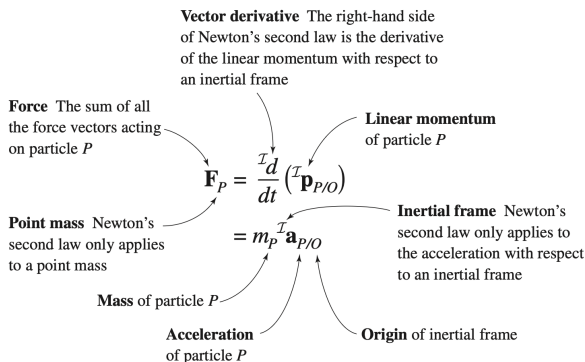


Figure 2: Anatomy of Newton's second law

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Definition

*A solution $\mathbf{r}(t)$ to newton's law is called a **motion**. The image of a motion is called a **trajectory**.*

Definition

Newton's law of universal gravitation states that each mass particle attracts every other particle in the universe with a force that varies directly as the product of the two masses and inversely as the square of the distance between them. In symbols,

$$\mathbf{F}_{i,j} = -G \frac{m_j m_i}{\|\mathbf{r}_i - \mathbf{r}_j\|^3} (\mathbf{r}_i - \mathbf{r}_j),$$

where $\mathbf{r}_i$ is the position of the i th particle, m_i its gravitational mass, and $\mathbf{F}_{i,j}$ indicates the force acting on particle i due to particle j .

Definition

- The **inertial mass** is that determining the acceleration of a body under the action of a given force.
- The **gravitational force** is that determining the gravitational forces between a body and other bodies.

The assertion of the exact equality between inertial and gravitational mass is termed the **principle of equivalence** and is considerably important in the general theory of relativity.

Example 1.5

Find the acceleration, velocity and position of a block sliding down a frictionless inclined (by an angle ϑ) fixed plane.

Theorem 1.6 (Conservation Theorem for the Linear Momentum of a Particle)

If the total force $\mathbf{F}_P$ is zero, then the linear momentum ${}^I\mathbf{p}_{P/O}$ is conserved.

Definition

The **angular momentum** of a particle about a point O , denoted by ${}^{\mathcal{I}}\mathbf{L}_{P/O}$, is defined as

$${}^{\mathcal{I}}\mathbf{L}_{P/O} = \mathbf{r}_{P/O} \times {}^{\mathcal{I}}\mathbf{p}_{P/O},$$

where $\mathbf{r}$ is the radius vector from O to the particle. The **torque** acting on point P about a point O is defined as

$$\mathbf{N}_{P/O} = \mathbf{r}_{P/O} \times \mathbf{F}_P.$$

Theorem 1.7 (Conservation Theorem for the Angular Momentum of a Particle)

If the total torque $\mathbf{N}_{P/O}$ is zero, then the angular momentum ${}^{\mathcal{I}}\mathbf{L}_{P/O}$ is conserved.

Remark

Henceforth, we will drop some notation whenever the reference frames and point are not specified, understood from the context or irrelevant.

Definition

Given a continuous vector field $\mathbf{F}$ and a path $\gamma : [a, b] \rightarrow \mathbb{R}^3$, we define the **vector line integral** to be the number

$$\int_{\gamma} \mathbf{F} = \int_a^b (\mathbf{F} \circ \gamma) \cdot \gamma'.$$

Definition

The **work** W done by an external force $\mathbf{F}$ over a path γ is defined as the line integral

$$W = \int_{\gamma} \mathbf{F}.$$

Example 1.8

Show that for constant mass, the work done by an external force reduces to

$$W = \frac{1}{2}m(\mathbf{v}_2^2 - \mathbf{v}_1^2).$$

Definition

*The scalar quantity $m\mathbf{\dot{r}}^2/2$ is called the **kinetic energy** of the particle and is denoted by T .*

Definition

A **retarding force** $\mathbf{F}_r$ is a *velocity-dependent force* $\mathbf{F}_r(\mathbf{v})$.

Remark

*Real retarding forces are complicated, but for the case of velocities that do not vary greatly (in norm), we can use the **power-law** approximation, which states that $\mathbf{F}_r \propto v^n$ with $v = ||\mathbf{v}||$. In this case we write*

$$\mathbf{F}_r = -mkv^n \frac{\mathbf{v}}{v},$$

*where the minus sign indicates that the force is always opposite to the velocity. The air resistance is known (experimentally) to be a retarding force, called the **drag force**. Experimentally, we find that, for a relatively small object moving in air, $n \approx 1$ for velocities about 24m/s, in this case the force is said to satisfy **Stoke's law of resistance** (it is directly proportional to the velocity). For higher velocities but below the velocity of sound $\sim 330\text{m/s}$, the drag force is approximately proportional to the square of the velocity.*

Definition

The **Prandtl expression** for the air resistance is

$$W = \frac{1}{2} c_W \rho A v^2$$

where c_W is a dimensionless drag coefficient, ρ the air density, v is the velocity, and A is the cross-sectional area of the object measured perpendicular to the velocity.

Example 1.9

As the simplest example of air resistance, consider a particle moving horizontally in a medium in which the drag force satisfies Stoke's law. Find the position and velocity as functions of t and furthermore, find the velocity as a function of the position.

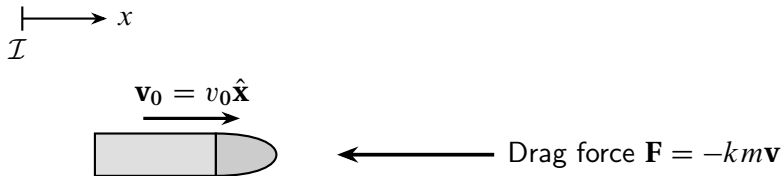


Figure 3: Example 1.9

Definition

If the force $\mathbf{F}$ is such that

$$\int_{\Gamma} \mathbf{F} = \int_{\Sigma} \mathbf{F}$$

for any two paths Γ, Σ joining the same initial and final points, then the force is said to be **conservative**. Using the Fundamental Theorem of Calculus for Line Integrals we have that such condition is equivalent to require that

$$\oint_C \mathbf{F} = 0$$

for all closed contours C , and also equivalent to require that there exists a scalar function V of $\mathbf{r}$ such that[†]

$$\mathbf{F} = -\nabla V(\mathbf{r}).$$

In this context, V is called the **potential**, or **potential energy**.

[†]The minus sign is customary and added due to physical reasons.

Definition

*The **total energy** E of a particle is defined to be the sum*

$$E = T + V.$$

Theorem 1.10 (Energy Conservation Theorem for a Particle)

If the forces acting on a particle are conservative, then the total energy E of the particle is conserved.

Definition

*The potential V is always assumed to be real-analytic, i.e., it can be approximated by a Taylor series around any point of its domain. We usually choose the point to be a relative extremum of V , such points are called **equilibrium points**.*

Definition

The equilibrium points are further characterised into three distinct types:

- i) **Stable** if the equilibrium point is a relative minimum of V .
- ii) **Unstable** if the equilibrium point is a relative maximum of V .
- iii) **Metastable** if it is neither stable nor unstable.

Example 1.11

Consider the one-dimensional potential

$$V(x) = -\frac{Wd^2 (x^2 + d^2)}{x^4 + 8d^4}.$$

Sketch the potential and discuss the possible motion of a particle under the influence of such potential.

Example 1.12

A particle of mass m moves under the influence of the potential

$$V(\mathbf{r}) = V_0 \left[\left(\frac{x}{a} \right)^2 + \left(\frac{y}{a} \right)^2 \right]$$

where V_0, a are positive constants. Sketch the potential and study the possible motions of the particle. In particular, study the case for $E = V_0$.

Remark

In generalising the ideas of the previous section to systems of many particles, we must distinguish between the external forces acting on the particles due to sources outside the system, and internal forces on, say, the i th particle due to all other particles in the system. Thus, Newton's second law for the i th particle is written as

$$\sum_j \mathbf{F}_{ji} + \mathbf{F}_i^{(e)} = \dot{\mathbf{p}}_i,$$

*where $\mathbf{F}_i^{(e)}$ stands for an external force, and $\mathbf{F}_{ij}$ is the internal force on the i th particle due to the j th particle. We shall assume that the $\mathbf{F}_{ij}$ (like $\mathbf{F}_i^{(e)}$) obey Newton's third law. This assumption is sometimes referred to as the **weak law of action and reaction**.*

Definition

If a frame $\mathcal{R}'$ moves with velocity ${}^{\mathcal{R}}\mathbf{V}$ relative to another frame $\mathcal{R}$, then the velocities ${}^{\mathcal{R}'}\mathbf{v}_j$ and ${}^{\mathcal{R}}\mathbf{v}_j$ of the particles are related through ${}^{\mathcal{R}}\mathbf{v}_j = {}^{\mathcal{R}'}\mathbf{v}_j + {}^{\mathcal{R}}\mathbf{V}$, and hence the momenta are related by

$${}^{\mathcal{R}}\mathbf{p} = {}^{\mathcal{R}'}\mathbf{p} + {}^{\mathcal{R}}\mathbf{V} \sum_j m_j.$$

The position of the reference frame $\mathcal{R}'$ in which the total momentum $\mathbf{p}'$ is zero is given by

$$\mathbf{R} = \frac{m_i \mathbf{r}_i}{\sum_j m_j}.$$

Definition

The position of the **centre of mass** $\mathbf{r}_{G/O}$ of a collection of N particles is defined as

$$\mathbf{r}_{G/O} = \frac{\sum_j m_j \mathbf{r}_j}{M}, \quad M \doteq \sum_j m_j.$$

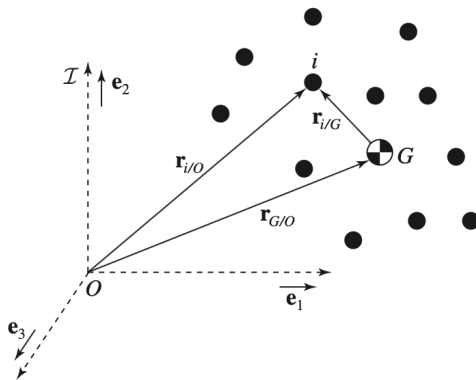


Figure 4: Centre of mass G of a collection of many particles

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